Algorithm:

Surface Blasting Design

Step 1: Begin.

Step 2: Declare variables Blasting Diameter(BD), Average In-Hole Density (p), Spacing To Burden Ration (a), Bench Height (BH), Stemming Length(SL), Burden(B), PFtech, Mass, and Actual Powder Factor(PFact).

Step 3: Input values for BD, p, a, PFtech, and BH.

Step 4: If the mine is located close to infrastructure, calculate SL by multiplying 30 by the BD, then dividing the product by 1000.

Step 5: Otherwise, if the mine is in a remote area, calculate SL by taking the product of 20 and BD, divide the product by 1000.

Step 6: Output the SL results.

Step 7: Calculate Mass by taking the product of p and the BD squared, divide the product by 1273. Output the Mass

Step 8: Calculate B by taking the square root of the product of the Mass and H, subtracting SL, divide the product by the product of a, PFtech, and BH.

Step 9: Output the B result.

Step 10: If B is between 3 and 3.9, calculate SD by taking the product of 1/3 and B.

Step 11: Calculate S by taking the product of a and B.

Step 12: Otherwise, if B is between 4 and 4.9, calculate SD by taking the product of ¼ and B.

Step 13: Output SD results

Step 14: If the rock is soft, calculate PFact by taking the product of M and BL, subtracting SL and SD, dividing the product by the product of B and S, and BH.

Step 15: Output the results of PFact.

Step 16: End

PSEUDOCODE

```
FUNCTION SurfaceBlastingDesign
    DECLARE BD, p, a, BH, SL, B, PFtech, Mass, PFact, S, SD
    DECLARE MineLocation, RockType
    INPUT BD
    INPUT p
    INPUT a
    INPUT PFtech
    INPUT BH
    INPUT MineLocation
    INPUT RockType
    IF MineLocation = "Close to Infrastructure" THEN
        SL = (30 * BD) / 1000
    ELSE
        SL = (20 * BD) / 1000
    END IF
    OUTPUT "Stemming Length =", SL
    Mass = (p * BD * BD) / 1273
    OUTPUT "Mass =", Mass
    B = SQRT((Mass * BH - SL) / (a * PFtech * BH))
    OUTPUT "Burden =", B
    IF B >= 3 AND B <= 3.9 THEN
        SD = (1 / 3) * B
    ELSE IF B >= 4 AND B <= 4.9 THEN
        SD = (1 / 4) * B
    END IF
    OUTPUT "Sub-drilling =", SD
    S = a * B
    IF RockType = "Soft" THEN
        PFact = (Mass * (BH - SL - SD)) / (B * S * BH)
        OUTPUT "Actual Powder Factor =", PFact
    END IF
END FUNCTION
```

FRAGMENTATION CALCULATOR

ALGORITHM

1. Start

2. Declare variables: In hole density(P), Bench height(BH), Relative Effective energy of explosive(E), Stemming length(SL), Hole diameter(D)

3. Ask the user to input the variables

4. If the rock is hard, use a powder factor of 0.7kg/m^3 and a rock constant(K) of 12-14.

- Otherwise if the rock is medium use a powder factor of 0.45kg/m^3 and a rock constant of 10-11.

- Otherwise if the rock factor is soft use a powder factor of 0.30kg/m^3 and a rock constant(K) of 8-9.

5. Calculate the mass of explosives using the formula M = $\frac{P \times D^2}{1273}$

6. Calculate charge length(CL) using the formula CL = $BH - SL$

7. Calculate the mass charge length(Q) using the formula Q = $M \times CL$

8. Verify if the variables in step 2 matches the one in the formulas

9. Calculate mean fragmentation size(X_50) = $A \times K^{-0.8} \times Q^{\frac{1}{6}} \left(\frac{115}{E}\right)^{\{0.633}$

10 End

PSEUDOCODE

BEGIN

```
DECLARE P, BH, E, SL, D
DECLARE M, CL, Q, X50
DECLARE A, K
DECLARE RockType

OUTPUT "Enter In-Hole Density (P):"
INPUT P

OUTPUT "Enter Bench Height (BH):"
INPUT BH

OUTPUT "Enter Relative Effective Energy (E):"
INPUT E
```

```
    OUTPUT "Enter Stemming Length (SL):"
    INPUT SL

    OUTPUT "Enter Blast Diameter (D):"
    INPUT D

    OUTPUT "Enter Rock Type (Hard, Medium or Soft):"
    INPUT RockType

    IF RockType = "Hard" THEN
        A = 0.7
        K = 13
    ELSE IF RockType = "Medium" THEN
        A = 0.45
        K = 10.5
    ELSE IF RockType = "Soft" THEN
        A = 0.30
        K = 8.5
    ELSE
        OUTPUT "Invalid Rock Type"
        STOP
    ENDIF

    M = (P × D²) / 1273
    CL = BH − SL
    Q = M × CL
    X50 ← A × K^(-0.8) × Q^(1/6) × (115 / E)^(0.633)
    OUTPUT "Mass of Explosives (M): ", M
    OUTPUT "Charge Length (CL): ", CL
    OUTPUT "Mass Charge Length (Q): ", Q
    OUTPUT "Mean Fragmentation Size (X50): ", X50
END
```